OncoEvidence: Mechanism-Guided Evidence Triage for Cancer Drug Repurposing

Knowledge-graph mechanism extraction and literature verification that is more precise than keyword screening and reproduces expert curation at scale

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The problem. Drug repurposing seeks new uses for existing drugs, for example whether an approved drug could treat a cancer it was not designed for. Models can rank thousands of drug-disease pairs, but a score alone is not actionable: a reviewer needs the *mechanism* (why the drug might work) and *evidence* (whether the literature supports it). OncoEvidence ranks oncology candidates, extracts the mechanism path that would explain each one, and uses a language model to verify that mechanism against published abstracts.

Thesis. The contribution is a *methodology comparison*, not a single model. *Mechanism-grounded evidence triage* (knowledge-graph mechanism extraction plus a literature-grounded language model) is measured against the two status-quo ways to vet repurposing candidates: a manual / expert-curated process, and naive automated screening. We separate *ranking* (where a tuned tabular model already wins) from *mechanistic justification* (where the graph is essential), which is the honest, defensible contribution.

- **More precise than keyword screening.** In a *small pilot* the language-model verifier made fewer but more precise "supported" calls than keyword / co-occurrence screening: **6/7** of its DrugMechDB-covered supported calls were curated-confirmed, versus **13/22** for keyword matching. The sample is small (wide confidence intervals), so we report this as a promising signal, not a final number.
- **Matches expert manual curation, automatically.** The extracted mechanisms agree with the expert-curated DrugMechDB at **0.802**, reproducing by pipeline what is normally hand-curated, across hundreds of pairs.
- **Honest baseline check.** On the plain link-ranking task a tuned tabular model (XGBoost) matches or beats the graph network, so the graph earns its place by producing the traceable *mechanism* a tabular model cannot, not by the score. Mechanism paths separate true from random indications at **AUROC 0.879**.

1 Background and Significance

Developing a new drug takes many years and large cost, while thousands of approved drugs already have known safety profiles. If an approved drug acts on a protein or pathway that also drives a cancer, it can be a fast route to a new treatment. Biomedical knowledge graphs such as PrimeKG encode millions of relationships among drugs, proteins, genes, pathways, and diseases, so the information needed to reason about such mechanisms exists as paths through the graph. The open problem is *trust*: a bare drug-disease score hides its reasoning, and cannot separate a real mechanism from a coincidence. OncoEvidence addresses this by making the mechanism explicit and checking it against the literature and a curated database.

Novelty, stated carefully. Knowledge-graph repurposing with explanations is not new: TxGNN is a graph foundation model with multi-hop rationales, and KGML-xDTD predicts treatment links with KG-based mechanism paths. We do not claim "a knowledge graph plus a language model for repurposing" as novel. Our narrower, defensible claim is methodological: we quantify *when* graph structure adds value beyond a strong tabular ranker, show that its value is mechanism extraction rather than better ranking, and then *audit* those mechanisms against curated MOA paths and sentence-level literature evidence to produce conservative, evidence-backed oncology dossiers.

2 Central Question and Specific Aims

Core question. For cancer drug repurposing, can we *separate ranking from justification*, using a strong non-graph ranker to prioritize candidates while using a biomedical knowledge graph and a literature-grounded language model to extract, audit, and score the mechanistic evidence behind each one, and what does the graph add once a tuned tabular ranker is already in place?

Specific Aims

Aim 1 (Candidate generation with calibrated baselines). Use a tuned XGBoost as the *primary* ranker, and treat the graph network and a knowledge-graph embedding as ablations, quantifying whether graph message passing improves ranking under leakage-safe transductive and inductive splits.

Aim 2 (Mechanism extraction, the graph's real role). The graph is not the best ranker; it is the *mechanism engine*. Extract multi-hop *drug* → *target* → *pathway/cancer-gene* → *cancer* paths, down-weighting promiscuous hubs.

Aim 3 (Evidence verification). A language model grades each path supported / weak / contradicted / unknown over retrieved abstracts, with a verbatim quote, treating a statistical association as weak rather than supporting.

Aim 4 (Evaluation, with hard controls). Separate true indications from *degree-matched and same-area hard negatives* (not only random pairs), measure agreement with curated DrugMechDB mechanisms, and report verifier precision with confidence intervals and leakage controls.

3 Datasets and System

Resource	Content	Role
PrimeKG (processed)	129,312 nodes; ∼8.1M *directed* edges over 50 directed relation types in our processed graph; 7,957 drugs, 17,080 diseases (3,170 oncology), 9,388 indications	Ranking + mechanism paths
Europe PMC	abstracts and open-access full text via REST	Evidence retrieval
DrugMechDB	curated drug mechanism-of-action paths (UniProt-keyed)	Mechanism ground truth

Table 1: Public resources used. PrimeKG supplies the graph; Europe PMC the evidence; DrugMechDB the curated mechanisms (mapped UniProt → HGNC via mygene.info). Note: the published PrimeKG release reports 129,375 nodes and 4,050,249 *undirected* relationships (10 node types, 30 edge types); our pipeline materializes directed reciprocal edges, which roughly doubles the edge count and relation-type count cited above.

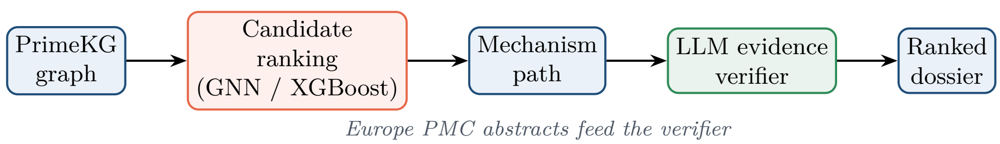

Figure 1: The OncoEvidence pipeline: rank candidates, extract a mechanism path, verify it against the literature, output an evidence-backed shortlist.

4 Methods

4.1 Candidate ranking and the honest baseline

Repurposing is framed as link prediction on *drug* → *indication* → *disease* edges. A heterogeneous graph neural network encodes each node by message passing over its typed neighbours; with a GraphSAGE-style layer,

$$h_v^{(l)} = \sigma\Big(W^{(l)}\big[\,h_v^{(l-1)} \;\big\|\; \mathrm{AGG}_{u\in\mathcal{N}(v)}\, h_u^{(l-1)}\big]\Big), \qquad s(d,c) = \sigma\big(z_d^{\top} R\, z_c\big), \tag{1}$$

where $s(d,c)$ scores a drug-disease pair. The graph network, a DistMult knowledge-graph embedding, and a tuned XGBoost baseline all receive the *same* SentenceTransformer node features, so any gap measures the value of graph structure, not features.

4.2 Mechanism-path extraction

For a candidate we search PrimeKG for short paths through mechanism relations (drug-target, protein-protein, protein-pathway, protein-disease), scoring each by specificity and penalizing promiscuous hubs:

$$\mathrm{spec}(x) = \frac{1}{\log_2(\deg(x)+2)}, \qquad \mathrm{score}(p) = \beta_{\mathrm{type}} + \mathrm{spec}(\mathrm{bridge}) \cdot \frac{1}{\log_2(\delta_{\mathrm{drug}}+2)}, \tag{2}$$

where δ_{drug} is the number of drugs that bind the bridge protein (an inverse-frequency penalty that suppresses generic carriers such as albumin), and β_{type} orders direct-target (3) over interaction (2) over shared-pathway (1) templates.

4.3 Evidence verification

For the top path we retrieve abstracts (a multi-query Europe PMC merge biased toward mechanism statements), extract the sentences mentioning both the drug and the target gene, and ask a language model to grade the mechanism as supported / weak / contradicted / unknown, requiring a verbatim quote and treating a purely statistical association as *weak*, not supporting.

4.4 Evaluation metrics

We report AUROC for separation, agreement with curated mechanisms, and the precision of "supported" verdicts:

$$\text{precision} = \frac{\#\{\text{“supported” pairs whose path gene is in the DrugMechDB MOA set}\}}{\#\{\text{“supported” pairs the DrugMechDB covers}\}}. \quad (3)$$

5 Preliminary Results

5.1 Finding 1: the link task does not need the graph

With a properly tuned XGBoost on shared features, the graph network does not win in any regime (Table 2). The earlier impression that it did was traced to an untuned baseline and a message-passing leak, both fixed. We report this plainly: for link ranking with rich text features, a strong tabular model suffices.

Test AUROC (5 seeds)	GNN	XGBoost	MLP	KGE
Transductive	0.977	**0.988**	0.973	0.907
Inductive (cold-disease, oncology)	0.882	**0.963**	0.930	0.452
Inductive (cold-drug)	0.956	**0.958**	0.931	0.482

Table 2: Corrected benchmark. A tuned XGBoost matches or beats the GNN everywhere; the DistMult memorizer (KGE) collapses on unseen nodes, confirming the task needs content features, not memorized identity.

5.2 Finding 2: the graph carries real mechanism

A tabular model cannot produce a traceable mechanism; the graph can. On true oncology indications the extractor recovers textbook direct-target mechanisms:

```
Quizartinib --targets--> FLT3 <--associated-- myeloid leukemia
Tamibarotene --targets--> RARA <--associated-- acute promyelocytic leukemia
Topotecan --targets--> TOP1 <--associated-- ovarian cancer
Trifluridine --targets--> TYMS <--associated-- colorectal cancer
```

Over 400 true indications vs 400 random drug-cancer pairs, the mechanism signal separates the two groups strongly (Table 3, Fig. 2).

Metric	True	Random
Mean mechanism score	**1.95**	0.17
Direct-target rate	**34.0%**	0.25%
Any mechanistic path	**81.3%**	8.8%

Separation AUROC
0.879

Table 3: Mechanism paths separate true indications from random pairs. A true indication is roughly 136× more likely to have a direct drug-target to cancer-gene link than a random pair.

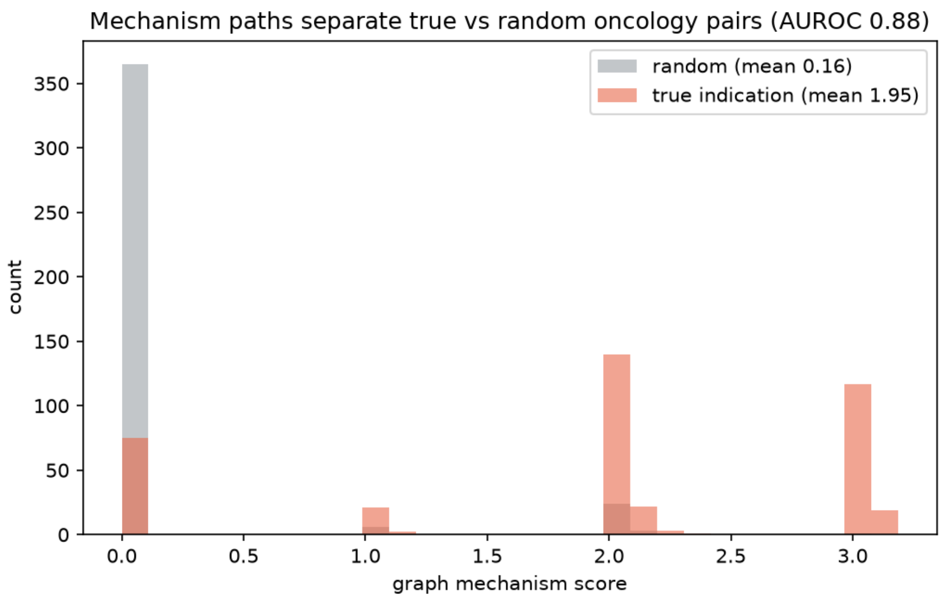

Figure 2: Distribution of the graph mechanism score for true indications (orange) vs random drug-cancer pairs (grey). Random pairs pile up at zero; true indications spread to high, specific scores (AUROC 0.879).

5.3 Finding 3: the mechanisms agree with a curated database

Mapping DrugMechDB's UniProt accessions to HGNC symbols makes a direct comparison possible. On the 263 true pairs DrugMechDB covers, our extracted bridge genes overlap the curated mechanism genes for **211 pairs (agreement 0.802)**.

Drug	Overlapping curated genes	
Methotrexate	ATIC, DHFR, TYMS	✓
Topotecan	TOP1	✓
Decitabine	DNMT1, DNMT3A	✓

Table 4: Examples of agreement with curated DrugMechDB mechanisms. DHFR is methotrexate's canonical target, recovered by the extractor.

5.4 Finding 4: the language model is a conservative, precise reviewer

Over 50 true vs 50 random pairs, the verifier marks far fewer "supported" than a keyword baseline, but those calls are more trustworthy (Table 5). A stricter rubric (an explicit drug-to-target statement is required; a genetic or statistical association is graded weak) plus sentence-level grounding collapsed "unknown" from 14 to 1. We report the raw counts, because the precision denominators are small.

	Language model	Keyword baseline
True pairs graded "supported"	11 / 50	34 / 50
Random pairs graded "supported"	1 / 50	2 / 50
"Supported" confirmed by DrugMechDB	**6 / 7**	13 / 22

Table 5: The language model is the conservative reviewer: of its "supported" calls that DrugMechDB covers, 6/7 are confirmed, against 13/22 for keyword matching. The denominators are small (Wilson 95% CI roughly 0.49–0.97 for 6/7 and 0.39–0.77 for 13/22), so this is a pilot signal, not a settled number, and the planned summer work scales it. Each verdict carries a verbatim quote, e.g. Topotecan / TOP1: *"a topoisomerase I (TOP1) inhibitor"*.

5.5 Robustness: hard negatives and a cold-start GNN test

Two stress tests, run after an external review, keep the framing honest.

Harder negatives. Replacing random negatives with progressively harder ones degrades the mechanism-score separation, as it should, but it stays above chance except in the hardest case (Table 6). Drugs that share a target or pathway with a true drug for that cancer yet lack the indication ("shared-target") are the genuinely hard case (AUROC 0.61). An ablation that removes *direct*-target paths still separates (AUROC 0.887), so the signal is not merely the most obvious drug-target link.

Negatives (vs true indications)	random	oncology-drug	degree-matched	shared-target
Separation AUROC	0.887	0.870	0.742	0.609

Table 6: Separation degrades with negative difficulty (400 vs 400, seed 0). The shared-target case is the honest hard limit. A no-direct-target ablation holds at AUROC 0.887, so indirect mechanisms also separate.

Does a cold-start GNN out-rank XGBoost under non-semantic features? No. Stripping name semantics (structural-only features, i.e. degree and node type, given identically to both models) in the cold-disease regime collapses XGBoost's text edge (AUROC 0.96 $\rightarrow$ 0.86), but a tuned XGBoost still narrowly out-ranks the GNN (0.86 vs 0.84, both seeds). A random-feature reference (GNN 0.84 $\approx$ structural-GNN 0.84) shows the achievable AUROC here reflects a drug-side identity prior, not message passing. We therefore do *not* claim a cold-start GNN ranking win; the graph's value stays in mechanism extraction and verification.

5.6 The graph's one genuine edge: blinded mechanism recovery

We trained the GNN jointly (link BCE plus a contrastive loss that ranks the curated DrugMechDB bridge gene above degree-matched decoys, using the same encoder), and asked it to *name the bridge gene* for held-out cold-disease pairs (3 seeds, $n \approx 186$–260). This is an axis a tabular model cannot touch: XGBoost never embeds a third (gene) node. Two readings, both honest (Table 7). *Unblinded*, a trivial "the drug's own targets" baseline dominates (Recall@10 0.80), because the curated bridge gene is usually just the drug's direct target; the GNN adds nothing there. But in the *mechanism-blinded* setting, with the direct drug→target edge removed, the target-lookup and the degree prior collapse to ~ 0, while the joint GNN still recovers the bridge gene (Recall@10 0.25 ± 0.02, per-seed 0.23–0.26, MRR 0.21, 3 seeds, $n \approx 186$–260) from indirect structure. Crucially, a link-only GNN scored through the *identical* mechanism head (same architecture and scorer, trained without the auxiliary loss) also recovers nothing (Recall@10 0.00), so the capability is attributable to the auxiliary objective, not to the architecture or a favorable scorer.

Bridge-gene recovery	Unblinded R@10	Blinded R@10	Blinded MRR
Joint GNN (link + mechanism)	0.31	**0.25**	**0.21**
Trivial target-lookup	**0.80**	0.00	0.00
Link-only GNN (same head, no aux loss)	0.00	0.00	0.00
Degree prior	0.02	0.02	0.02

Table 7: Mechanism recovery on held-out cold-disease pairs. Unblinded, the bridge gene is the drug's direct target, so a trivial lookup wins. When that direct edge is hidden, only the joint GNN recovers the gene via indirect structure, a small (∼0.25 R@10) but genuine graph-only signal. The link-only GNN uses the same head and scorer, differing only in the auxiliary loss. Caveats: DrugMechDB supervision is drug-level (the drug's MOA genes), not pair-specific; even blinded, the bridge gene may remain reachable through retained PPI/pathway edges, so this reflects recovery of *known* mechanism via indirect paths, not de novo discovery; and $n \approx 200$ covered held-out pairs is modest, so it is a preliminary signal to scale. A real but narrow advantage, not a blockbuster.

6 Discussion: what the method beats (and what it does not)

The contribution is best read as a comparison of three ways to vet cancer repurposing candidates. Against *keyword / co-occurrence screening*, the language-model verifier is more precise in the pilot (6/7 confirmed vs 13/22; denominators small), because it reads whether the text actually states a mechanism rather than whether the words merely co-occur (a keyword screen grades a pharmacogenetic association as "supported"; the language model correctly downgrades it to weak). Against the *manual curation* process, the pipeline reproduces expert DrugMechDB mechanisms at 0.802 automatically and at scale, doing in seconds per candidate what manual literature review does by hand. We are explicit about what the method does *not* beat: on the plain link-ranking task a tuned tabular model is as good as the graph, so we do not claim a model win there; the graph earns its place by producing the traceable mechanism that a tabular model cannot.

Limitations, stated plainly. We do not claim to beat human experts on *quality*; we match curated quality while automating and scaling it. The separation partly reflects that drugs indicated for a cancer already have a known target linked to that cancer in the graph, so this validates that the graph encodes *known* mechanism consistently, rather than discovering new biology. Some metrics rest on modest samples (the verifier precision on seven covered "supported" pairs) and partial coverage (DrugMechDB; the open-access full-text subset). The planned summer work scales the verifier run, broadens full-text coverage, and applies the same triage to *novel* candidate pairs to generate mechanism-verified repurposing hypotheses.

7 Eight-Week Timeline

Weeks	Milestones
1–2	PrimeKG pipeline; leakage-safe splits; shared features.
3–4	Candidate ranking + tuned XGBoost baseline (Finding 1).
5	Mechanism extraction with hub down-weighting (Findings 2–3).
6	Literature retrieval and the verifier (Finding 4).
7	Full evaluation; scale the verifier; error analysis.
8	Deliverable shortlist, figures, written report.

Table 8: Weeks 1–6 are already de-risked by the preliminary results above.

8 Reproducibility and Ethics

Open tools (PyTorch Geometric, scikit-learn, XGBoost) and public data (PrimeKG, Europe PMC, DrugMechDB); code, configs, seeds, and the UniProt→HGNC cache are released, and every reported number is multi-seed where applicable. All predictions are research-only and hypothesis-generating, not medical advice.

References

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